PROBLEM CL20

Pipe Contraction Discharging to the Atmosphere: Velocity, Force and Heating

Original statement · CL20.pdf

Nozzle | Linear momentum | Energy | Bernoulli limit

OFFICIAL PROBLEM STATEMENT

A liquid of density ρ and specific heat c flows steadily along a circular pipe of diameter D, discharging to the atmosphere, at pressure p_a, through a coaxial contraction that ends with a circular orifice of diameter d. Upstream from the contraction, the velocity, pressure, and temperature have known uniform values v_u, p_u, and T_u. As a function of these data (ρ, c, D, d, p_a, v_u, p_u, T_u), obtain explicit expressions for:

- The value of the discharge velocity v_d.
- The force acting on the contraction F, neglecting in the calculation the effect of gravity.
- The temperature increment $T_d - T_u$. Assume in the calculation that the contraction wall is adiabatic.

Simplify these last two results by further assuming that the effect of viscosity is entirely negligible, so that for this incompressible steady flow Bernoulli's equation can be used to relate the variation of the fluid properties along a given stream line.

Source: CL20.pdf (page 1, final of Jan 21, 2014). Notes.pdf: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid, $e = cT$; neglect $\bar{g}$; uniform properties at the upstream section u and at the orifice d; adiabatic contraction; free jet at p_a.
- Data: $[\rho,\ c,\ D,\ d,\ p_a,\ v_u,\ p_u,\ T_u]$. Unknowns: $[v_d,\ \bar{F},\ T_d - T_u]$, first in general and then in the inviscid (Bernoulli) limit.

Symbol	Meaning	SI unit
D, d	pipe and orifice diameters	m

Symbol	Meaning	SI unit
v_u, p_u, T_u	upstream state	m/s, Pa, K
v_d, T_d	discharge velocity and temperature	m/s, K
$\bar{F}$	force on the contraction	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the contraction, $\bar{e}_x$ along the axis. Fixed control volume: the liquid inside the contraction.
- Surfaces: Σ_u (upstream section, $\bar{n} = -\bar{e}_x$, area $\pi D^2/4$), Σ_d (orifice, $\bar{n} = +\bar{e}_x$, area $\pi d^2/4$), Σ_w (contraction wall, outer face at p_a).

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

1. Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v}\cdot\bar{n}\, d\sigma$
Σ_u	$-\rho v_u \frac{\pi D^2}{4}$
Σ_d	$+\rho v_d \frac{\pi d^2}{4}$
Σ_w	0

$$\boxed{v_d = \left(\frac{D}{d}\right)^2 v_u} \tag{1}$$

2. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \bar{v}\, dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho \bar{v}\,[(\bar{v} - \underbrace{\bar{v}_c}_{0}) \cdot \bar{n}]\, d\sigma = -\int_{\Sigma_c} (p - p_a)\,\bar{n}\, d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma + \underbrace{\int_{V_c} \rho \bar{g}\, dV}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \bar{v}\,(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int (p-p_a)\,\bar{n}\, d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\, d\sigma$
Σ_u	$-\rho v_u^2 \frac{\pi D^2}{4}\bar{e}_x$	$(p_u - p_a)\frac{\pi D^2}{4}\bar{e}_x$	0
Σ_d	$+\rho v_d^2 \frac{\pi d^2}{4}\bar{e}_x = \rho v_u v_d \frac{\pi D^2}{4}\bar{e}_x$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\, d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\, d\sigma$

Boxed terms: $-\bar{F}$. The Σ_d flux uses (1).

$$\rho\frac{\pi D^2}{4} v_u (v_d - v_u)\,\bar{e}_x = (p_u - p_a)\frac{\pi D^2}{4}\bar{e}_x - \bar{F} \implies \boxed{\bar{F} = \frac{\pi D^2}{4}\left\{p_u - p_a - \rho v_u^2\left[\left(\frac{D}{d}\right)^2 - 1\right]\right\}\bar{e}_x} \tag{2}$$

3. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\, d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\, d\sigma + \underbrace{\int_{V_c} Q_H\, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	$-\int \bar{q}\cdot\bar{n}\, d\sigma$
Σ_u	$-\dot{m}\left(cT_u + \frac{v_u^2}{2}\right)$	$+p_u v_u \frac{\pi D^2}{4}$	0	0

Surface	$\int \rho \left(e + \frac{v^2}{2}\right)(\bar{v} \cdot \bar{n})\, d\sigma$	$-\int p\, \bar{v} \cdot \bar{n}\, d\sigma$	$\int \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma$	$-\int \bar{q} \cdot \bar{n}\, d\sigma$
Σ_d	$+\dot{m}\left(cT_d + \frac{v_d^2}{2}\right)$	$-p_a v_d \frac{\pi d^2}{4}$	0	0
Σ_w	0	0	0	0 (adiabatic)

With $\dot{m} = \rho v_u \pi D^2/4 = \rho v_d \pi d^2/4$, dividing by $\dot{m}c$:

$$\boxed{T_d - T_u = \frac{1}{c}\left[\left(\frac{p_u}{\rho} + \frac{v_u^2}{2}\right) - \left(\frac{p_a}{\rho} + \frac{v_d^2}{2}\right)\right]} \quad (3)$$

4. Inviscid limit (Bernoulli)

Without viscosity, Bernoulli (Notes Eq. 4.17) along a streamline from u to d gives $\frac{p_u}{\rho} + \frac{v_u^2}{2} = \frac{p_a}{\rho} + \frac{v_d^2}{2}$, i.e. $p_u - p_a = \frac{1}{2}\rho v_u^2 \left[\left(\frac{D}{d}\right)^4 - 1\right]$. Substituting into (2), with $\left(\frac{D}{d}\right)^4 - 2\left(\frac{D}{d}\right)^2 + 1 = \left[\left(\frac{D}{d}\right)^2 - 1\right]^2$, and into (3):

$$\boxed{\bar{F} = \frac{\pi D^2}{8}\rho v_u^2 \left[\left(\frac{D}{d}\right)^2 - 1\right]^2 \bar{e}_x, \qquad T_d - T_u = 0} \quad (4)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- (3): the temperature rise measures the loss of mechanical energy (total pressure) by viscous dissipation; without viscosity it is zero, (4).
- In the inviscid limit $\bar{F}$ points downstream and is always positive: the contraction is pulled along the flow. For $d \to D$, $\bar{F} \to 0$.
- Dimensions (SI): $[\rho v^2 D^2] = \mathrm{N}$; $[(p/\rho + v^2/2)/c] = \mathrm{K}$.
- Numerical illustration (hypothetical data): water, $D = 0.1$ m, $d = 0.04$ m, $v_u = 1$ m/s, $p_u = 1.21 \times 10^5$ Pa, $p_a = 101325$ Pa: $v_d = 6.25$ m/s, $F = 113.29$ N and $T_d - T_u = 0.154$ mK (some loss); the inviscid value is $F = 108.24$ N (it needs $p_u - p_a = 19031$ Pa).
- Agreement with the official solution: CL20.pdf obtains (1) to (4).